

Research paper

A novel development of h-CSLQFL controller for optimal sizing and economic pricing of battery in electric vehicle system

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ABSTRACT

This paper proposes a novel design of h-CSLQFL controller for reducing the battery size (kWh) and battery cost (US\$/kWh) of electric vehicle (EV) system. The h-CSLQFL controller is a hybrid controller which is formed by the combination of Cuckoo search optimization (CSO), linear quadratic regulator (LQR), and fuzzy logic controller (FLC). The higher battery cost is one of the biggest challenges with EV due to which a significant number of consumers cannot afford to purchase it. The replacement of conventional vehicles with EV creates ecofriendly and sustainable solutions. For attaining the optimal reduced cost of battery, a strategy has been demonstrated by framing a multi-objective function which is controlled with h-CSLQFL controller. It commences with the design of mathematical model of EV in which supply voltage and current are the input parameters whereas the speed and torque are the output parameters. Thereafter, LQR controller was designed and implemented for analyzing the performance parameter of EV. Afterwards, CSO controller in support with LQR controller has been formulated in closed loop manner. At the end, the multi-objective function is controlled with LQR based CSO and LQR controller. It is observed that the least optimal economic cost and size of EV battery have been attained with h-CSLQFL controller in comparison to the LQR based CSO and LQR controller. In addition to this, the least risk factor and reduced sensitivity have been also attained with the proposed approach. The validation of the proposed h-CSLQFL controller has been done on a hardware set up.

1. Introduction

With the passing time, the fatalistic result of fossil fuel has been experienced in transportation systems and their negative impact on the augmentation of the greenhouse effect. Keeping these worse effects into consideration, the need for reducing carbon and saving energy is a matter of concern now. These effects will impact the market of vehicles working on the internal combustion engine because of the rising price of fossil fuels. The different technologies of electric vehicles (EVs) have been developed and advanced to get a solution to these (energy and environmental) problems (Haddoun et al., 2008). The system's modelling, analysis, and simulations of an electric vehicle with two independent rear wheel drives are presented in the work. In the absence of differential gears, the traction control system is intended to ensure the acceleration and stability of the electric vehicle. Each wheel can be

equipped with torque and speed control by using two in-wheel electric motors. The above setting enhances the safety and stability of EVs. The vehicle velocity, which is distinct from the steering wheel speed indicated by slipping in the operating mode, is the data input used by the suggested traction management system. To predict the vehicle speed in this scenario, a modified neural network technique is suggested.

Contemplating resources of energy and protection of the environment from the harsh effect of ordinary fuel sources, EVs having better performance because of several advantages such as smooth operation, improved efficiency is taking place in the market. Additionally, EVs are showing a better role in fuel economy improvement (Khooban et al., 2016). Internal combustion engine vehicles (ICEVs) and EVs have differences in their power system. In EVs, the power system electrification brings new sound quality (SQ) problems and challenges into the light. The noise of electric motors is one of the main noise sources in EVs along

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with other noise sources like a wide range of speed, load current harmonics, and large electromagnetic loads (Ma, 2017). When it comes to designing EVs, some performance parameters are desired such as high-power density, a wide range of speed regulation, low noise, strong overload capability, and high efficiency. Recently, stator-permanent magnet flux-switching (SPM-FS) machines were developed because of having better torque capability and better operation reliability. But this also resulted in many challenges to overcome (Urquhart et al., 2013).

Compared to combustion engines, EV motors have a flexible layout. Among different powertrain structures of EVs, distributed electric powertrain (DEP) is special. In DEP, each driving wheel is equipped with an independent powertrain. Different strategies are proposed to enhance the energy consumption efficiency of DEP (Hederic and Hadžiselimovic, 2014). It has led to the production of EVs at a huge rate and many developments are taking place for the improvement of the same. Many charging facilities have been constructed for EVs and a huge number of EV charging stations (EVCSs) are being planned and designed keeping future scope and expansion into consideration (Ghavami et al., 2016). The main power source of a hybrid electric vehicle (HEV) is gasoline, and the secondary power source is a high-voltage battery, and its development and production are very rapid as it causes negligible harm to the environment causing low pollution. In Japan and the U.S., the HEVs are already being mass-produced for general consumers. The advantages of HEV are that they are compatible with existing powertrains because of having similar components such as a motor, an engine, and a transmission. Dual-clutch transmission and continuously variable transmission have been selected as the transmission of HEVs (Amir et al., 2021).

Permanent magnets are getting more value in the automobiles industry because of having high power density and great robustness. In these machines, the need for maintenance is less because rotor magnets replace field winding, and this also increases the overall efficiency. By improving the mechanical structure or by using advanced controlling methods, the power of PM machines can be improved significantly. To optimize the PM machines, it requires a focus on proper magnet shape and winding configuration (Haque and Rahman, 2009).

The stability of automobiles is one of the main focuses of study these days. It is very important to design and develop reliable brakes. In this, the internal winding of one motor is separated into two parts namely, the master part and the slave part, and connected to two EV-control units (ECUs). This turned out to be an advantage for the system but also brought the possibility of overheating which may lead to the burning of both the parts because of the flow of the same current through both of the parts (Boyuan et al., 2017). In plug-in electric vehicles, instead of having a voluminous charging circuit having many huge numbers of inductors, propulsion circuit components are used for charging the battery. This leads to the minimization of weight as well as cost savings. In this system, the motor winding is used to replace the grid interfacing inductance and the propulsion converter is utilized as a dc or ac converter, which is called on-board integrated battery chargers (Hagbhin et al., 2013).

The contraption of smart charging and demand-side management is a very salient need. Receding-horizon optimization for valley filling has been proposed to schedule EV charging. But only the EV which is already connected to the power grid is being considered in this. A new method, which is a prediction-based real-time smart charging method (N-RT) is recently introduced which is based on the real-time decentralized charging (RT) (Ma et al., 2013). Keeping the progress in the field of EV automobile industry, it can be said that due to residential EV charging, electric power consumption is going to rise in the coming years. In the study, the probability of an EV to be parked at residence is analysed, availability was found maximum during the night, and between 10 AM to 6 PM, the average availability was found to be 0.7 approximately. And this helped in building a voltage support method that was based on storage solutions (Carvalho et al., 2008). A game theory strategy has been used to assess the charge control of EV battery (Kim et al., 2022). In

this, an EV regularly determines whether to charge its battery in a distributed way based on the data. In order to prevent an EV from leaving a charging station with inadequate energy, a restricted probabilistic game model is developed, and its solution is found using a method that utilizes optimal reaction dynamic. The goal is to reduce the typical expense of charging an EV. In comparison to a threshold-based charge scheme, the assessment findings demonstrate that D-EVCM can lower the average charge cost by over 48 % given a low enough battery outage probability.

The charge scheduling and optimal cost of battery of EV is discussed in (Xie et al., 2020). Plug-in electric car parking lot equilibria with energy and reserve markets along with the optimal behaviour of electric vehicle parking lots as demand response aggregation agents has been shown in (Neyestani et al., 2015; Shafie-Khah et al., 2016; Almasri et al., 2024; Al-fouzan and Almasri, 2024). Allocation of plug-in vehicles parking lots in distribution systems taking network-constrained objectives into consideration is shown in (Neyestani et al., 2017). In (Shafie-Khah et al., 2018), an illustration of a ground-breaking two-level model for electric car parking spaces has been provided in renewable energy distribution systems. A clear explanation of the complexity analysis of the ideal charging schedule for EVs may be found in (Rottondi et al., 2016). In Monteiro et al. (2016), the current and future functioning modes for electric vehicles has been discussed in smart grids and smart homes. An implementation of machine learning for the fast charging of electric vehicle using renewable energy sources for grid integrated residential loads has been discussed in (Khezri et al., 2023). Further, a brief discussion on time of tariff for the electric vehicle owner using renewable energy sources like solar photovoltaic cell and battery energy storage system has been implemented in (Merrington et al., 2022). In the next study (Khezri et al., 2022), impact of wind energy and energy storage system on grid integrated based residential loads has been assessed in the presence or absence of electric vehicle.

The comparison of the proposed method with existing literatures have been analysed and mentioned in Table 1 in which various parameters like modelling, no. of variables, possible battery arrangement, electric vehicle battery size and cost, have been compared by considering the proposed method, LQR (Neyestani et al., 2017), machine learning algorithm (Khezri et al., 2023), voltage support method (Carvalho et al., 2008). In addition to this, hardware implementation has been compared for various methods. It is observed that proposed hybrid intelligent controller uses simple modelling with less no. of variables which gives least battery size and least battery cost in comparison to the existing methods. Such positive and dominant results of proposed hybrid intelligent controller show its novelty and advantageous over other controllers.

After performing the exhaustive literature survey, the most current problems like large battery size (kWh), high battery cost (US\$/kWh), and high risk factor and poor sensitivity of electric vehicle (EV) system have been found out. The complex modelling of electric vehicle system along with more number of variables and high potential difficult are also the few concern of existing literatures. Such issues have been addressed and overcome by the proposed h-CSLQFL controller which is the need of the study. It begins with modelling of EVs, subsequently LQR, CSO, LQR based CSO controller has been designed and developed for more analysis. A hybrid controller known as h-CSLQFL has been designed by combining the LQR, CSO, and FLC controller. At the end, performance analysis for the optimal size, cost, risk analysis and sensitivity of EV battery has been estimated with LQR, LQR based CSO and h-CSLQFL controller. It is observed that least optimal pricing, reduced size, least risk analysis, and improved sensitivity of EV battery have been attained with h-CSLQFL controller in comparison to LQR based CSO and LQR controller.

The paper is organized as follows: Section 2 presenting the mathematical model of the system and the design of the Linear Quadratic Regulator (LQR) controller. Further, Section 3 describing the Cuckoo Search Optimization (CSO) algorithm and its application in the

Table 1

Comparison of proposed method with existing control techniques.

Parameters/ Methods	LQR (Neyestani et al., 2017)	Machine learning algorithm (Khezri et al., 2023)	Voltage support method (Carvalho et al., 2008)	Proposed hybrid intelligent (h-CSLQ FL) controller
Modelling	Complex	Moderate	Moderate	Simple
No. of variable	More	Medium	Medium	Less
Possible battery arrangement	Less	Moderate	Moderate	High
EVs battery size	High	Moderate	Moderate	Less
EVs battery cost	High	Moderate	Moderate	Less
Hardware implementation	No	No	No	Yes
real-time processing	No	No	Mo	Yes
Potential difficulties	High	Tough	Tough	Moderate

controller design. Furthermore Section 4 proposing the integration of CSO with the LQR controller for improved performance. Also, Section 5 introducing the Hybrid Controller (h-CSLQFL) and its design process. Section 6 formulating the multi-objective function to estimate the optimal cost of the EV system. Section 7 evaluating the performance parameters of the system using the hybrid controller and comparing it with other methods. Section 8 shows the validation of proposed hybrid intelligent controller on hardware set up, Section 9 summarizing the findings of the study and discussing the implications of the proposed h-CSLQFL controller.

2. Mathematical modelling and design of linear quadratic regulator

The mathematical modelling and optimal design of Linear Quadratic Regulator (LQR) systems prove indispensable for the optimization of control strategies in the context of linear time-invariant systems.

2.1. Control design of EV drive system

The drive system of an EV is essential for creating a mathematical model. The actuator of an EV is a DC motor. Eq.1 expresses the relationship between torque and armature current since the torque produced by a DC motor is directly proportional to the armature current.

$$T_d = K_m I_a \quad (1)$$

Motor constant K_m is motor winding construction dependent parameter. Voltage developed in motor can be estimated as in Eq.2.

$$V_d = K_m \omega_d \quad (2)$$

where ω_d is armature rotational speed. Terminal voltage of motor can be estimated from Eq.3 as V_h .

$$V_h = I_h R_a + L_h \frac{di(t)}{dt} + V_d \quad (3)$$

where L_h is the inductor on the higher side, I_h is the terminal current at the higher side, and R_a is the armature's resistance.

T_d is comparable to output mechanical torque T_{mec} when inertial losses and friction are negligible. Therefore, the mechanical power produced equals the electrical power produced. As a result of Eq. 4, the high side input voltage can now be changed in terms of the low side voltage.

$$V_h = K V_L \quad (4)$$

High side input is calculated in terms of low side current as:

$$I_h = \frac{1}{K} I_L \quad (5)$$

where V_L and I_L are low side (output side) voltage and current and K presents controller's gain factor. As discussed earlier, the battery is a

voltage source with internal resistance R_{s_int} and open circuit voltage (V_{oc}), Now.

$$V_L = I_L R_a + E_b \quad (6)$$

In this work PI controller has been used to control charging mechanism of battery and feedback to PI controller is error of battery voltage calculated as.

$$V_{b(err)} = V_{oc(actual)} - V_{oc(calculated)} \quad (7)$$

So, motor constant K can be calculated with proportional and integral constant of PI controller.

$$K = (K_p + sK_i) V_{b(err)} \quad (8)$$

So required control pulses are generated and controlled with the help of PI controller.

2.2. Design of LQR

The linear quadratic regulator (LQR) technique is used to automatically improve the performance of the controller since it is a feedback controller employed as the ideal control system for any system with dynamic behavior (Zhang et al., 2022). When modelling and designing EVs, LQR is used to optimize the controller unit that controls speed, yaw, and lateral motion. It can also be used to improve power train control by using a kalman filter to detect battery open circuit voltage (Neyestani et al., 2017) and to regulate the regenerative braking system.

The mathematical design of LQR is represented from Eq.9 to Eq.15. LQR state space modeling of EV in terms of voltage, current and angular speed is presented with relation, $\dot{x} = Ax + Bu$ with a quadratic cost function defined as (9).

$$J = x^T(t_1)F(t_1)x(t_1) + \int (x^T Q_x + u^T R_u + 2x^T N_u) dt \quad (9)$$

The feedback control law that optimizes the value of the cost is $u = -Kx$, where K is given by $K = R^{-1}B^T$. It is found by solving the continuous-time Riccati differential equation:

$$A^T P + PA - PBR^{-1}B^T P + Q = 0 \quad (10)$$

with the boundary condition: $P(t_f) = F(t_f)$, the reference values of speed (ω) and current (i) can be solve with the LQR equations by matrix X as stated below:

$$X = \begin{bmatrix} i \\ \omega \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (11)$$

Error can be generated as ΔX by subtracting the signal fed from output side and reference signal applied to the system, hence can be calculated as:

$$\Delta X = \begin{bmatrix} \Delta i \\ \Delta \omega \end{bmatrix} \quad (12)$$

Now quadratic cost function for LQR technique is calculated using

error signal and LQR weight matrices as:

$$J = \int_0^{\infty} (\Delta X(t)^T Q \Delta X(t) + (\Delta U(t)^T R \Delta U(t)) dt \quad (13)$$

where $Q = \begin{bmatrix} Q_1 & 0 \\ 0 & Q_4 \end{bmatrix}$ is state cost weighted matrix and $R = [R_1]$ is input cost weighted matrix. After solving the $\frac{dJ}{dt} = 0$,

$$Q_1(\Delta i)^2 + Q_4\Delta\omega^2 + R_1(\Delta V)^2 = 0$$

$$V = I_a R_a + E = I_a R_a + K\omega$$

$$\Delta V = \Delta I_a R_a + K\Delta\omega$$

$$Q_1(\Delta i)^2 + Q_4\Delta\omega^2 + R_1(\Delta I_a R_a + K\Delta\omega)^2 = 0$$

$$Q_1(\Delta i)^2 + Q_4\Delta\omega^2 + K^2(\Delta\omega)^2 + R_a^2(\Delta I_a)^2 + 2K\Delta I_a R_a \Delta\omega = 0 \quad (14)$$

After differentiating Eq. (14) with respect to Δi , $\Delta\omega$ and $\Delta i \Delta\omega$, these methods are used to determine the value of K , or feedback gain, which is determined by solving the Riccati equation in algebra.

$$K = \begin{bmatrix} \frac{1}{I_a + I_f} & 0 \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \end{bmatrix} \quad (15)$$

The structure of LQR controller and its illustration in EV are shown in Fig. 1 and Fig. 2 respectively.

3. Design of Cuckoo Search Optimization (CSO)

A family of birds known as cuckoos has a distinctive and aggressive reproductive strategy when compared to other bird species. Some species of cuckoo birds, such as Ani and Guira, deposit their eggs in communal nests, however they sometimes remove the eggs of other cuckoo bird species to enhance the likelihood that their own eggs will hatch. Other species lay their eggs in the nests of other birds or in host nests using the brood parasitism technique. Parasitic cuckoos are adept at snatching nests where eggs have recently been laid, and they have

highly accurate timing for egg-laying. In the host nest, they lay one egg, which typically hatches more quickly than the other eggs. When this occurs, the alien cuckoo would push the unhatched eggs outside of the nest and remove them. This activity aims to lessen the likelihood that the genuine eggs will hatch. Additionally, by imitating the host chicks' calls, the alien cuckoo chick can get more food (Lu et al., 2022; Zhang et al., 2021). Sometimes the host cuckoo will notice that one of the eggs is not its own. The cuckoo then either discards the egg or leaves the nest entirely and moves to create a new nest somewhere else.

3.1. Lévy flights

The step lengths of Lévy flights, which are random walks with random directions, are derived from the Lévy distribution. Animals and insects conduct these Lévy flights, which are characterized by a sequence of straight flight paths followed by abrupt 90-degree rotations. Lévy flights are more effective at scouring vast search areas than regular random walks. This is mostly caused by the variances of Lévy flights increasing significantly more quickly than those of a typical random walk. Compared to a regular random walk, Lévy flights can cut the number of optimization algorithms iterations by around 4 orders.

3.2. CS algorithm

The cuckoo search (CS) algorithm was created using cuckoo bird reproduction as inspiration. It is crucial to relate potential solutions to cuckoo eggs while working with CS algorithms. In the hopes that their offspring will be nurtured by substitute parents, cuckoos typically lay their fertilized eggs in the nests of other cuckoos. Sometimes cuckoos find eggs in their nests that don't belong to them; in these situations, they either throw the foreign eggs out of the nests or abandon the entire nest.

Yang and Deb used three idealized rules for CS based on the brood parasitic behavior of a cuckoo:

- Each cuckoo only lays one egg at a time, which is then put in several nests at random.

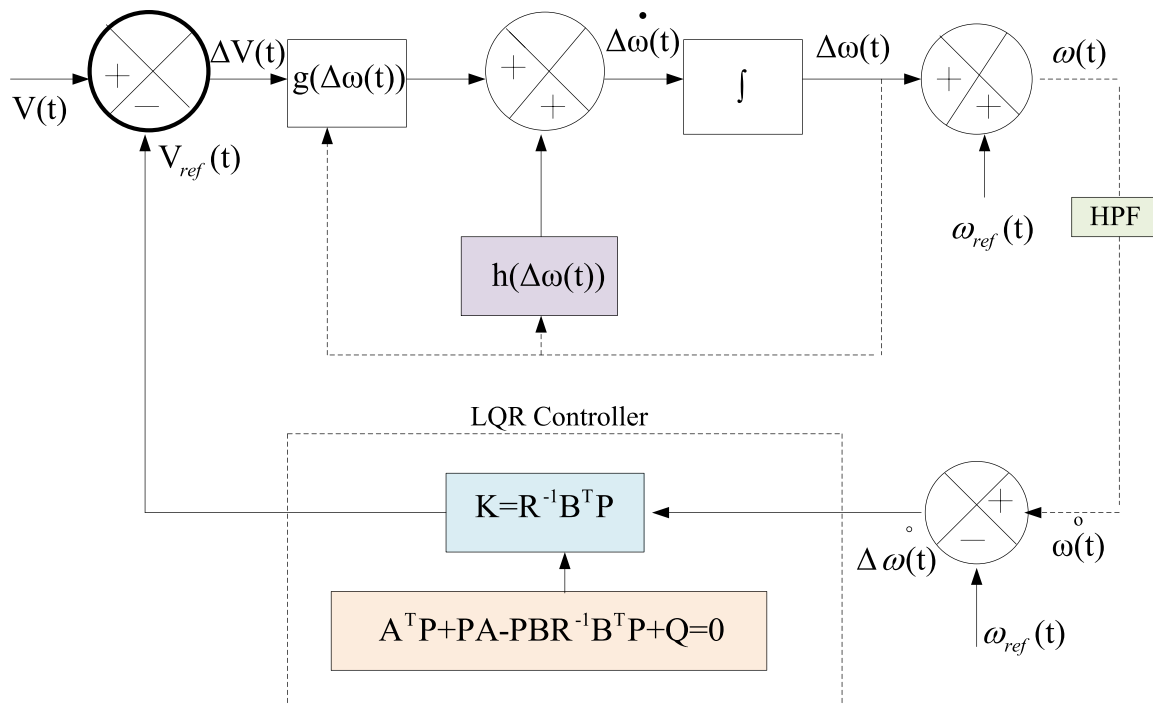


Fig. 1. Structure of LQR controller.

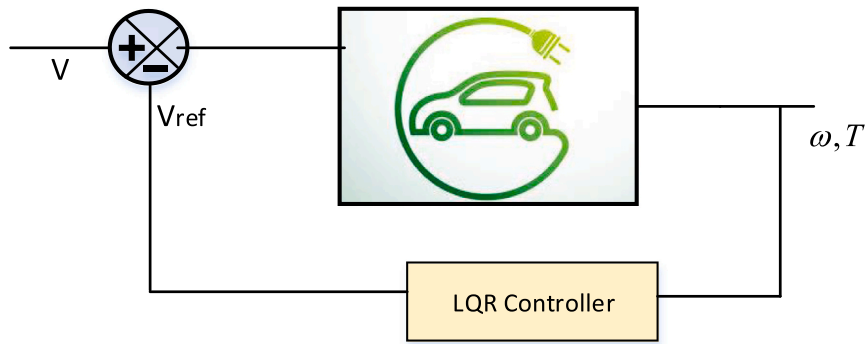


Fig. 2. Illustration of LQR in Electric vehicle.

- The best nest will produce the best eggs, which will be passed down to the following generation.
- The number of available nests is fixed, and the number of available nests (laid by a cuckoo) discovered by the host bird maintains a probability P_a , where $P_a \in [0,1]$.

On the basis of these three rules, the strategy of CS for a nest is as follows:

$$x_i^{t+1} = x_i^t + \alpha \otimes \text{Levy}(s, \lambda) \quad (16)$$

where x_i^t refers to position/egg sample, i is the sample number, s is step size, $\text{Levy}(s, \lambda)$ is Lévy distribution, used to define the step size of random walk and t is the number of iterations. The product ($\otimes$) indicates entry-wise multiplication, where $\alpha = \alpha_0(x_{cgb} - x_i^{t-1})$ simplified scheme of levy distribution is presented in as:

$$\text{Levy}(\lambda) \approx \frac{u}{(|v|)^{1/\beta}} \quad (17)$$

where $\beta = 1.5$, x_{cgb} is best position/nest, α_0 is the levy multiplying coefficient (chosen 0.8), whereas u and v are determined from the normal distribution curves, namely: $u \approx N(0, \sigma_u^2)$ and $v \approx N(0, \sigma_v^2)$

If Γ denotes integral gamma function, then the variables σ_u and σ_v are defined as:

$$\sigma_u = \left(\frac{\Gamma(1+\beta) \times \sin(\pi \times \beta/2)}{\Gamma\left(\frac{1+\beta}{2}\right) \times \beta \times 2^{\left(\frac{\beta-1}{2}\right)}} \right)^{1/\beta}, \sigma_v = 1 \quad (18)$$

The stride is made up of a number of small steps, as well as occasionally larger steps and long jumps, according to Levy distribution. In a low-power zone, the random step may obviously result in excessively small steps, and the zone previously searched by other cuckoos with small steps may be searched again, extending the tracking time. The process of CSO algorithm is explained through flowchart as shown in Fig. 3.

4. Design of LQR based CSO

The extended version of the LQR structure, which was covered in Section 3, is the LQR-based CSO. Fig. 4 depicts the main configuration of a LQR-based CSO for controlling EV. A comprehensive design overview of the LQR-based CSO controller is given in Fig. 5. The change in speed and the rate of change in speed are inputs to the CSO in this extended LQR controller architecture.

The first input of CSO is change in speed which is represented as A as shown in Eq.19.

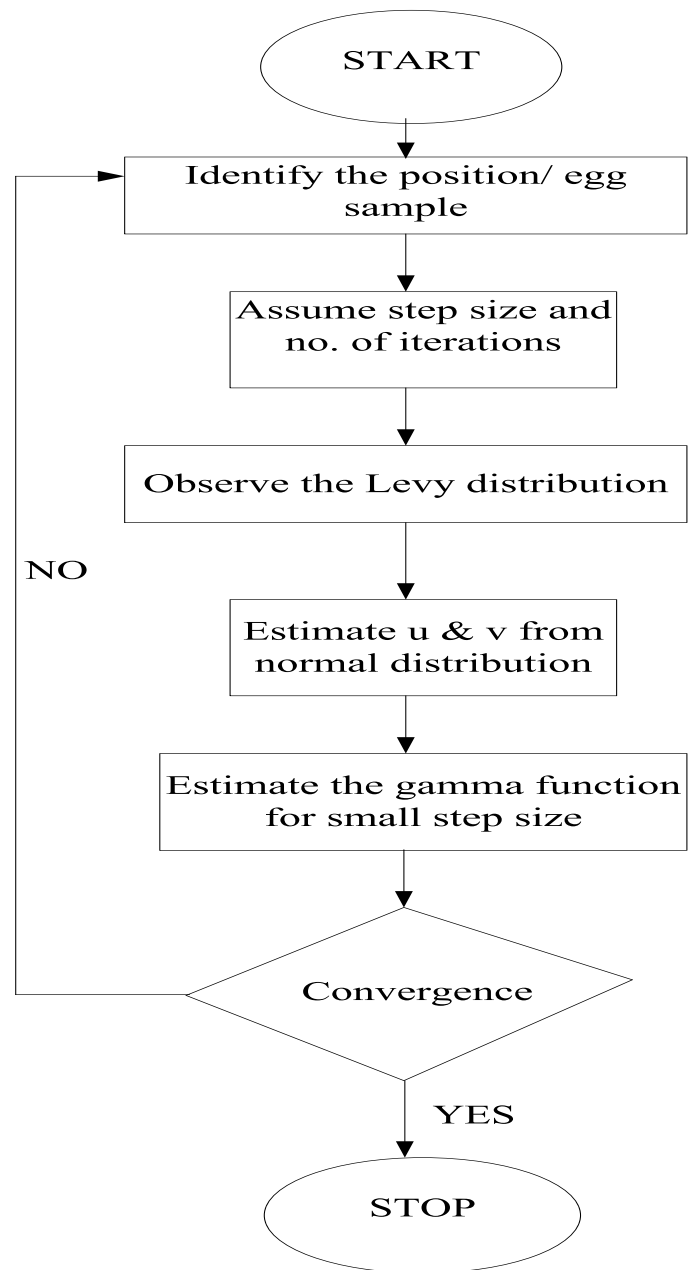


Fig. 3. Flow chart of cuckoo search optimization algorithm.

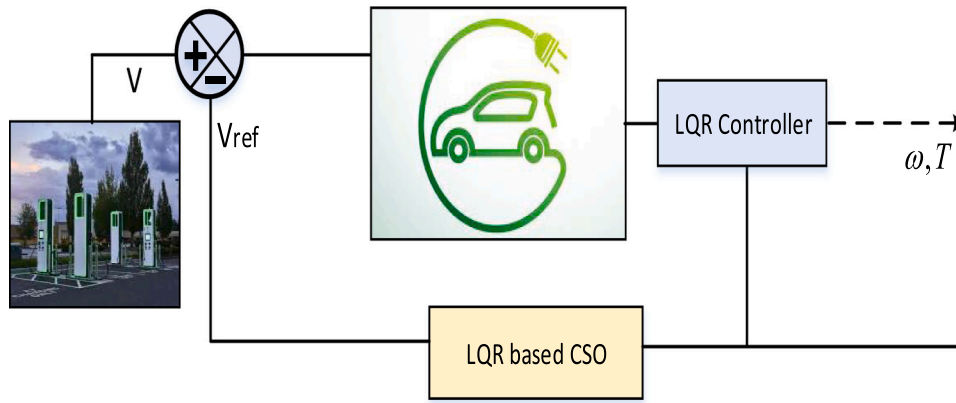


Fig. 4. Layout of LQR based CSO controller for electric vehicle.

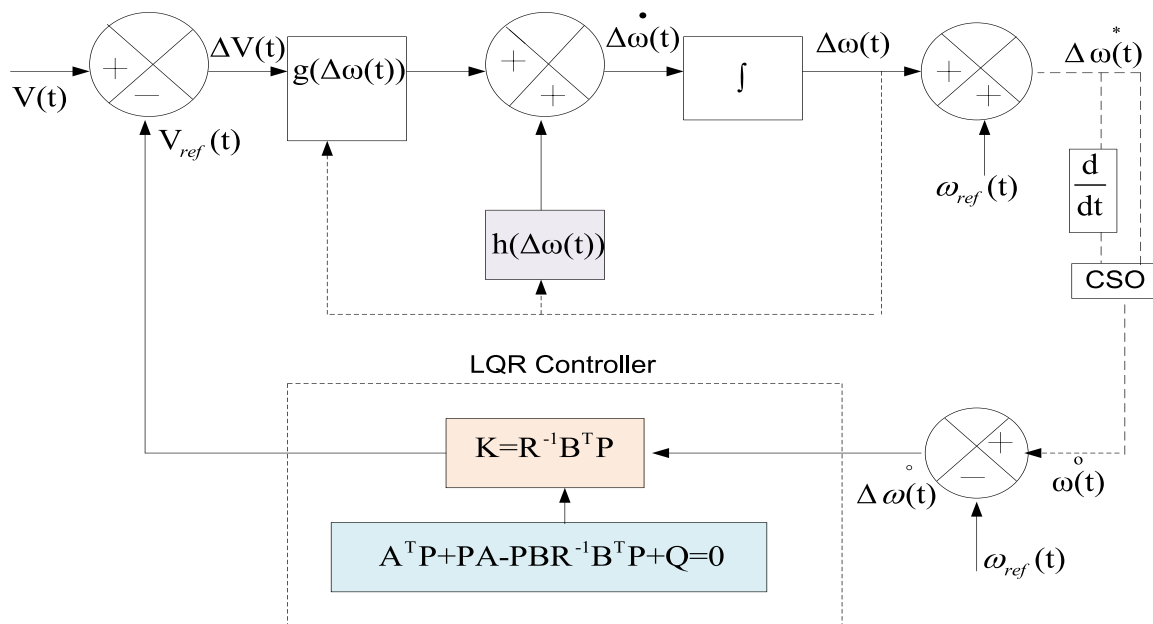


Fig. 5. Structure of LQR based CSO structure.

$$A = \Delta \omega = \omega_{ref}(t) - \Delta \omega(t) \quad (19)$$

Another input of CSO is the rate of change in speed which is represented as B as shown in Eq.20.

$$B = \frac{d(\Delta \omega)}{dt} \quad (20)$$

The CSO output is generated speed ($\Delta \omega(t)$) which is shown in Eq.21.

$$\Delta \omega(t) = \omega(t) - \omega_{ref}(t) \quad (21)$$

5. Design of hybrid controller (h-CSLQFL controller)

The design of h-CSLQFL controller is cumulative combination of CSO, LQR, and FLC controller. It begins with LQR based CSO controller. which further extends with the support of FLC for the modeling of hybrid controller. The model of hybrid controller has been designed by extending the Fig. 5, in which speed ($\omega(t)$) is being expressed in the form of error (e) by comparing with its delayed value as depicted in Eq.22.

$$e(t) = \omega(t) - \omega(t-1) \quad (22)$$

This is one of input of the FLC and another input will be taken as change in error as shown in Eq.23.

$$\Delta e(t) = e(t) - e(t-1) \quad (23)$$

These two inputs are trained with the help of certain set of rules and output of FLC is $\Delta \omega(t)$. The structure of the hybrid controller (h-CSLQFL Controller) representing the above mathematical approach is shown in Fig. 6.

The proper set of rules have been framed between two inputs for getting the proper optimal output. With the help of intersection topology between two inputs of the cramer's rule, 7×7 fuzzy rules have been designed between two inputs. The notation of fuzzy rules between two inputs has usual meaning as shown in Table 2.

6. Formulation of multi-objective function for estimation of optimal cost of electric vehicle

The multi objective function of EV battery charging which depends on battery size and battery cost is given by Eq.24.

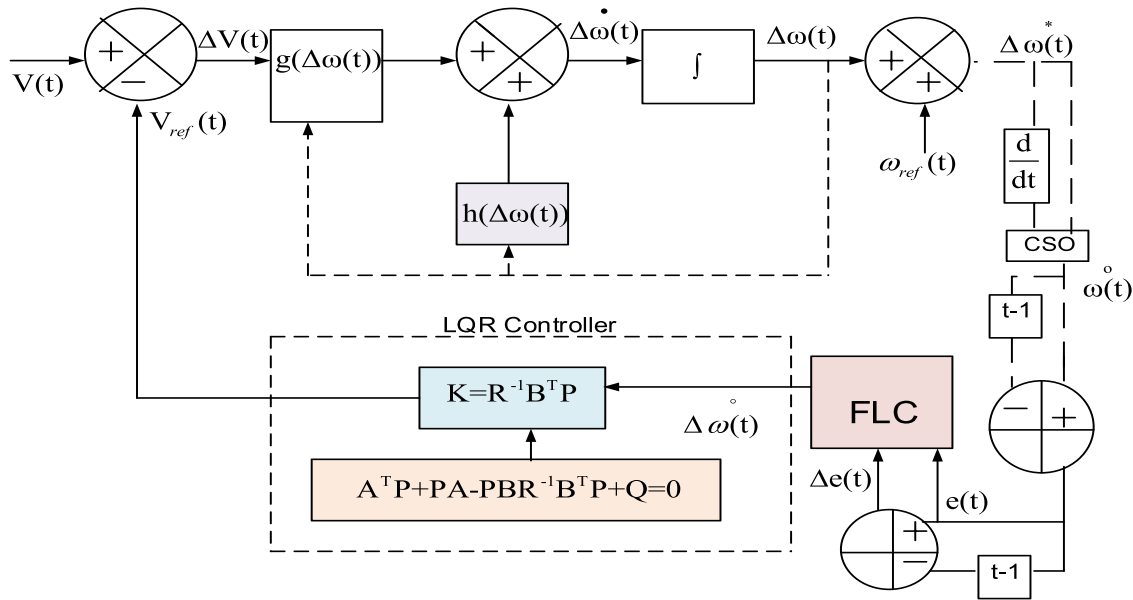


Fig. 6. Structure of hybrid controller (h- CSLQFL Controller).

Table 2
Fuzzy rules between two inputs.

Rules	NS	NM	NB	ZS	PS	PM	PB
NS	PB	PS	PM	ZS	NS	NM	NB
NM	NB	NM	NS	ZS	NB	NM	NS
NB	PM	NS	NB	ZS	NS	NB	PB
ZS	ZS	ZS	ZS	ZS	ZS	ZS	ZS
PS	PM	PB	PM	ZS	PM	PS	PB
PM	NB	NM	NS	ZS	PB	PM	PS
PB	NB	NM	NS	ZS	PB	PB	PM

Now the mathematical formulation of objective function will be developed for reducing the cost of EV battery which will be discussed in the next section.

$$f(x(t)) = f(BS(t), n, BP(t)) \quad (24)$$

Where, $BS(t)$ = battery size, n = no. of batteries

$BP(t)$ = Battery price, $SOC = P_{Solar} - P_{Delivery}$

$$SOC_{min}(t) < SOC(t) < SOC_{max}(t) \quad (25)$$

Another objective function for battery selection management and its size is given in Eq.26.

$$f(BS(t), n) = E * N - \frac{Qik_1N^2}{Q - it} - \frac{Qitk_2N^3}{Q - it} - \frac{k_3E}{N^4} \quad (26)$$

Under the constraint as given below:

$$k_1N + k_2N + k_3E < \frac{SOC(t)}{Q} \quad (27)$$

$$SOC(t) < SOC(t-1) - \frac{it}{Q} \quad (28)$$

It is needed to include the constraint the parameters from Eq.27 and Eq.28 in Eq.26, then Eq.29 is attained as:

$$f(BS(t), n) = E * N - \frac{Qik_1N^2}{Q - it} - \frac{Qitk_2N^3}{Q - it} - \frac{k_3E}{N^4} + c_1(k_1N + k_2N + k_3E - \frac{SOC(t)}{Q}) + d_1(SOC(t) - SOC(t-1) + \frac{it}{Q}) \quad (29)$$

In the Eq.29, inequality variables are c_1 and d_1 which is

mathematically represented as given in Eq.30 and Eq.31.

$$c_1 = e^{\frac{k_1N - k_2N}{k_3E}} \quad (30)$$

$$d_1 = e^{\frac{SOC(t) - SOC(t-1)}{Q}} \quad (31)$$

For getting the optimal solution, differentiate with Eq.29 respect to charge Q and equate to 0 as given in Eq.32.

$$\frac{d(f(BS(t), n))}{dQ} = 0 \quad (32)$$

$$Q = \frac{it}{1 \pm \sqrt{\frac{(itk_2N^3 - ik_1N^2)it}{SOC(t)c - d_1it}}} \quad (33)$$

Eq.33 is the two optimal solution of Eq.32 which gives the suitable value of charge. The optimal value of charge depends on the cost battery coefficient.

where E is battery size in kWh, n is the no. of batteries which are arranged in horizontal and vertical manner. k_1 , k_2 , k_3 are the cost coefficient of the battery (US\$/kWh) obtained after training with various methods.

Now, it is observed that optimal value of charges is giving approximately the same results as obtained for $g(\Delta\omega(t))$ and $h(\Delta\omega(t))$. It means two optimal solutions can be represented as in Eq.34 and Eq.35.

$$g(\Delta\omega(t)) = \frac{it}{1 + \sqrt{\frac{(itk_2N^3 - ik_1N^2)it}{SOC(t)c - d_1it}}} \quad (34)$$

$$h(\Delta\omega(t)) = \frac{it}{1 - \sqrt{\frac{(itk_2N^3 - ik_1N^2)it}{SOC(t)c - d_1it}}} \quad (35)$$

Now after obtaining the $g(\Delta\omega(t))$ and $h(\Delta\omega(t))$, it is convenient to apply hybrid controller, LQR based CSO, and LQR controller for obtaining the optimal solution.

The cost aspects of battery can be expressed in Eq.36.

Table 3

Estimation of Battery Size $f(BS(t), n, N^*)$ /Kwh with Different Arrangement of Battery using Various Controller Methods.

n	h -CSLQFL	LQR based CSO	LQR
1×8	39.8	41	43.7
8×1	39.2	40.4	43.1
4×2	40.1	41.3	44
4×4	38.9	40.1	42.8
2×4	39.9	41.1	43.8

$$BP(t) = k_4(E * N - \frac{Qik_1N^2}{Q - it} - \frac{Qitk_2N^3}{Q - it} - \frac{k_3E}{N^4}) + k_5(\frac{BP_{\max}(t) - BP_{\min}(t)}{n}) + k_6(\frac{SOC_{\max}(t) - SOC_{\min}(t)}{n}) \quad (36)$$

k_4 , k_5 , and k_6 are the cost factor in US\$/kWh under the constraint as

$$h(\Delta\omega(t)) = \frac{it}{1 - \sqrt{\frac{i^2tk_1N^2 - i^2t^2k_2N^3}{\&p_1(k_1N^2 + k_2N^3 + k_3E - \frac{BP_{\text{optimal}}(t)}{SOC_{\text{optimal}}(t)}) + q_1(BP_{\min}(t) - BP(t)) + r_1(BP_{\max}(t) - BP(t))}}} \quad (45)$$

$$g(\Delta\omega(t)) = \frac{it}{1 + \sqrt{\frac{i^2tk_1N^2 - i^2t^2k_2N^3}{\&p_1(k_1N^2 + k_2N^3 + k_3E - \frac{BP_{\text{optimal}}(t)}{SOC_{\text{optimal}}(t)}) + q_1(BP_{\min}(t) - BP(t)) + r_1(BP_{\max}(t) - BP(t))}}} \quad (46)$$

shown below in Eq.37 and Eq.38.

$$k_1N^2 + k_2N^3 + k_3E < \frac{BP_{\text{optimal}}(t)}{SOC_{\text{optimal}}(t)} \quad (37)$$

$$BP_{\min}(t) < BP(t) < BP_{\max}(t) \quad (38)$$

In order to avoid constraints, it has to be included in main equations by considering the inequality constraints as shown in Eq.39.

$$BP(t) = k_4(E * N - \frac{Qik_1N^2}{Q - it} - \frac{Qitk_2N^3}{Q - it} - \frac{k_3E}{N^4}) + k_5(\frac{BP_{\max}(t) - BP_{\min}(t)}{n}) + k_6(\frac{SOC_{\max}(t) - SOC_{\min}(t)}{n}) + \frac{p_1}{Q}(k_1N^2 + k_2N^3 + k_3E - \frac{BP_{\text{optimal}}(t)}{SOC_{\text{optimal}}(t)}) + \frac{q_1}{Q}(BP_{\min}(t) - BP(t)) + \frac{r_1}{Q}(BP_{\max}(t) - BP(t)) \quad (39)$$

Where, p_1 , q_1 , r_1 are the cost factors in US\$/kWh which can be obtained by various topologies and mathematically can be expressed as in Eqs.40, 41, and 42.

$$p_1 = e^{-\frac{k_1N^2 - k_2N^3}{k_3E}} \quad (40)$$

$$q_1 = e^{-\frac{BP_{\min}(t)}{BP(t)}} \quad (41)$$

$$r_1 = e^{-\frac{BP(t)}{BP_{\max}(t)}} \quad (42)$$

For getting the optimal solution, differentiate Eq.39 with respect to charge Q and equate to 0 as given in Eq.43.

$$\frac{d(f(BS(t), n))}{dQ} = 0 \quad (43)$$

There will be two optimal solutions obtained for the two values of Q which are mentioned as in Eq.44,

$$Q = \frac{it}{1 \pm \sqrt{\frac{i^2tk_1N^2 - i^2t^2k_2N^3}{\&p_1(k_1N^2 + k_2N^3 + k_3E - \frac{BP_{\text{optimal}}(t)}{SOC_{\text{optimal}}(t)}) + q_1(BP_{\min}(t) - BP(t)) + r_1(BP_{\max}(t) - BP(t))}}} \quad (44)$$

Similar type of observation has been assessed for the two values of Q as attained previously, which reflects that optimal values of Q are approximately closer to $g(\Delta\omega(t))$ and $h(\Delta\omega(t))$. The mathematical notation can be expressed as in Eq.45 and Eq.46.

After attaining the modified values of $g(\Delta\omega(t))$ and $h(\Delta\omega(t))$, the proposed hybrid controller, LQR based CSO, and LQR controller can be applied for analysing the performance parameters of EVs. The overall multi-objective function consisting of battery size, battery cost, SOC is represented in Eq.47.

$$f(x(t)) = f(BS(t), n, BP(t)) = E * N - \frac{Qik_1N^2}{Q - it} - \frac{Qitk_2N^3}{Q - it} - \frac{k_3E}{N^4} + c_1\left(k_1N + k_2N + k_3E - \frac{SOC(t)}{Q}\right) + d_1\left(SOC(t) - SOC(t-1) + \frac{it}{Q}\right) + k_4\left(E * N - \frac{Qik_1N^2}{Q - it} - \frac{Qitk_2N^3}{Q - it} - \frac{k_3E}{N^4}\right) + k_5\left(\frac{BP_{\max}(t) - BP_{\min}(t)}{n}\right) + k_6\left(\frac{SOC_{\max}(t) - SOC_{\min}(t)}{n}\right) + \frac{p_1}{Q}\left(k_1N^2 + k_2N^3 + k_3E - \frac{BP_{\text{optimal}}(t)}{SOC_{\text{optimal}}(t)}\right) + \frac{q_1}{Q}(BP_{\min}(t) - BP(t)) + \frac{r_1}{Q}(BP_{\max}(t) - BP(t)) \quad (47)$$

7. Performance parameters estimation using hybrid controller

The battery specification is measured in kWh making selection of available total battery is 8 in various arrangement for the estimation of battery size $f(BS(t), n)$ with various arrangement of battery is shown in Table 3 and its graphical comparison is shown in Fig. 7.

The overall comparative analysis for estimation of battery size in kWh for the various arrangement of battery reveals that least size of battery has been attained with h-CSLQFL controller in comparison to LQR based CSO and LQR controller for 4×4 arrangement. Similarly, the cost estimation of EV battery in US\$/kWh for the various possible arrangement has been reflected in Table 4 and its graphical comparison

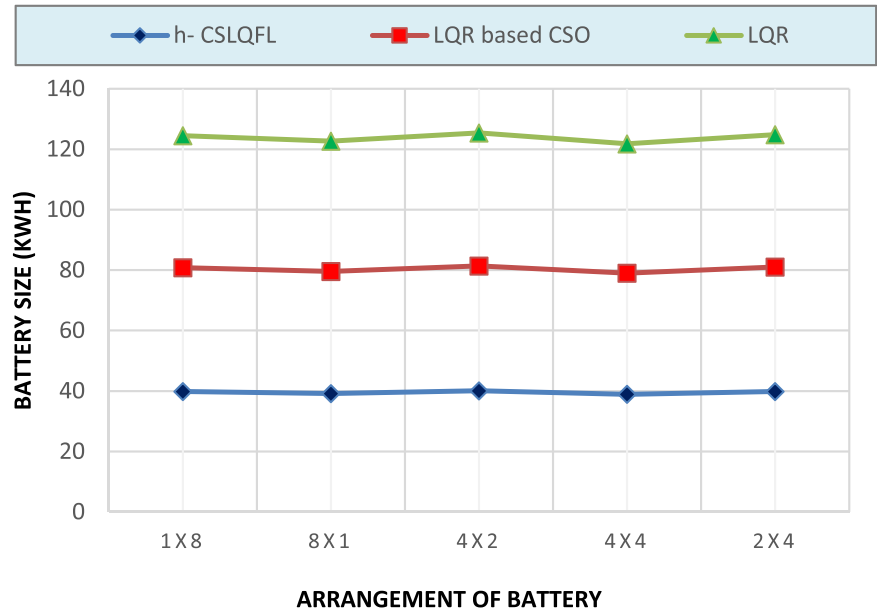


Fig. 7. Comparative analysis of battery size with various methods.

Table 4
Estimation of Electric Vehicle Cost ($BP(t)$) In Dollar/Kwh with Different Arrangement of Battery using Various Control Methods.

N	h- CSLQFL	LQR based CSO	LQR
1×8	1.22	1.51	1.91
8×1	0.9	1.12	1.52
4×2	1.01	1.21	1.61
4×4	0.84	0.99	1.39
2×4	1.21	1.35	1.75

Table 5 Risk Factor Analysis.				
Different loading condition	n	h-CSLQFL	LQR based CSO	LQR
25 % of full load	1×8	0.78	0.81	0.85
40 % of full load	8×1	0.68	0.7	0.72
60 % of full load	4×2	0.66	0.68	0.71
80 % of full load	4×4	0.32	0.45	0.51
full load	2×4	0.48	0.51	0.56

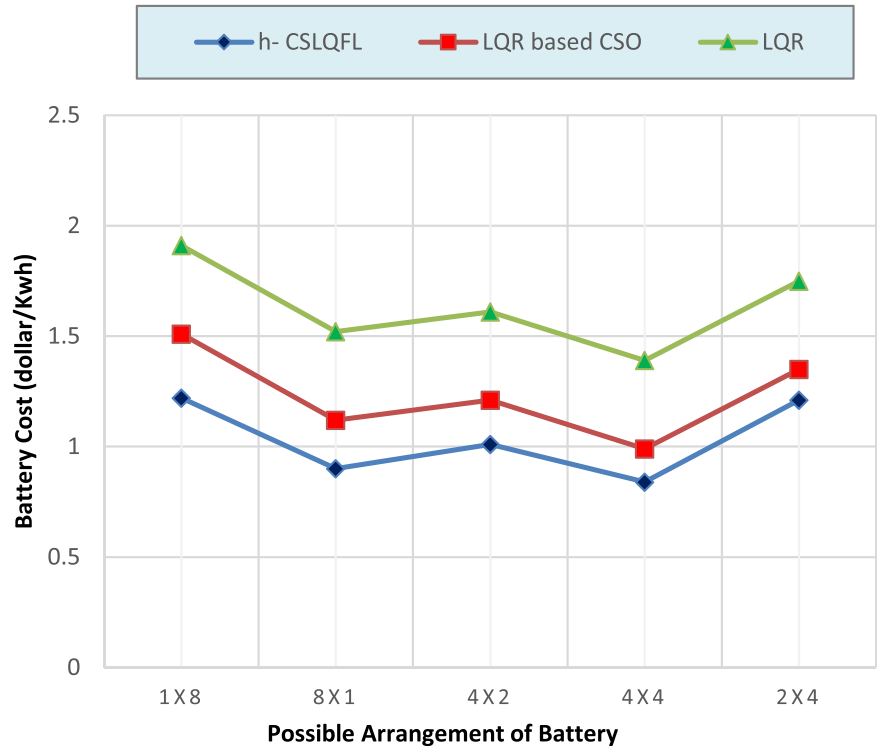


Fig. 8. Comparative analysis of battery cost in US\$/kWh with various methods.

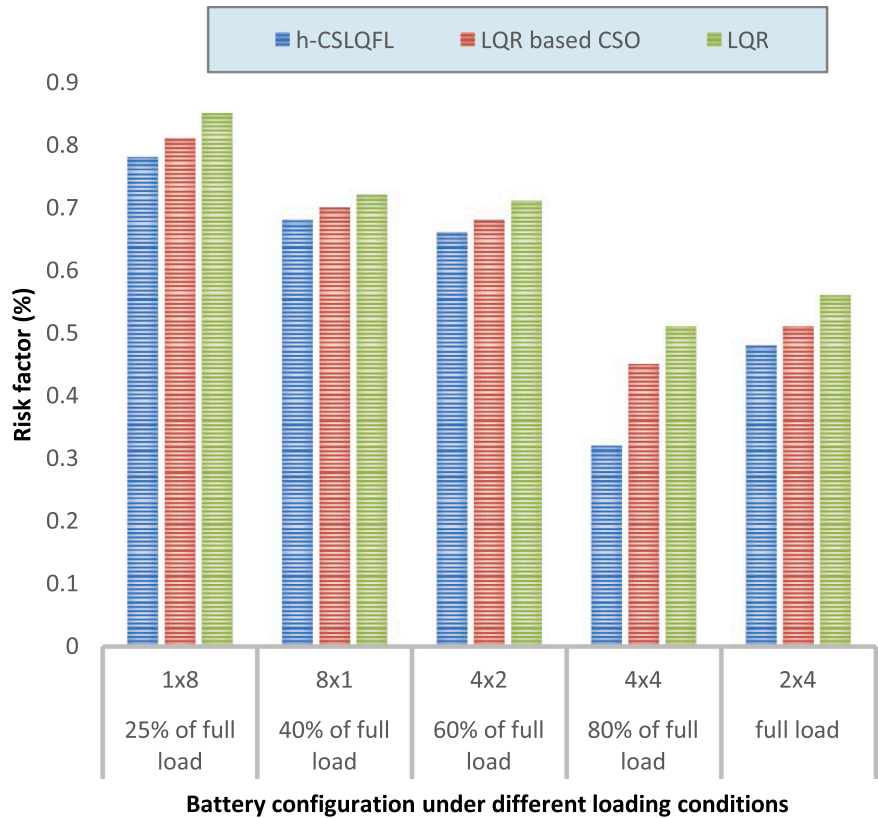


Fig. 9. Comparison of risk factor for various battery load configurations.

Table 6
Sensitivity analysis of different arrangement.

N	h- CSLQFL	LQR based CSO	LQR
1×8	0.89	1.35	1.69
8×1	0.78	1.45	1.69
4×2	0.86	1.26	1.84
4×4	0.91	1.68	1.91
2×4	0.85	1.31	1.87

is shown in Fig. 8. It has been observed that least cost of battery has been acquired with h-CSLQFL controller in comparison to LQR based CSO and LQR controller for 4×4 arrangement.

After the estimation of battery cost and its size, it is required to

measure the life expectancy of the battery which is measured in terms of risk factor as represented in Eq.48.

$$\text{Risk factor} = \frac{\min \text{ desired KWH} * f(\text{BS}(t), n)}{\text{Rated KWH} * \text{BP}(t)} * N \tag{48}$$

Such analysis of risk factors under different loading conditions with hybrid controller, LQR based CSO controller, LQR controller is shown in Table 5.

Using a 4×4 arrangement, the least amount of risk has been achieved with hybrid controller (h-CSLQFL) in comparison to other methods. It can be concluded that configuration of battery with a 4×4 structure can be considered as best because it is least expensive, smallest in size, least risk factor with hybrid controller as shown in Table 3 and its graphical comparison is shown in Fig. 9.

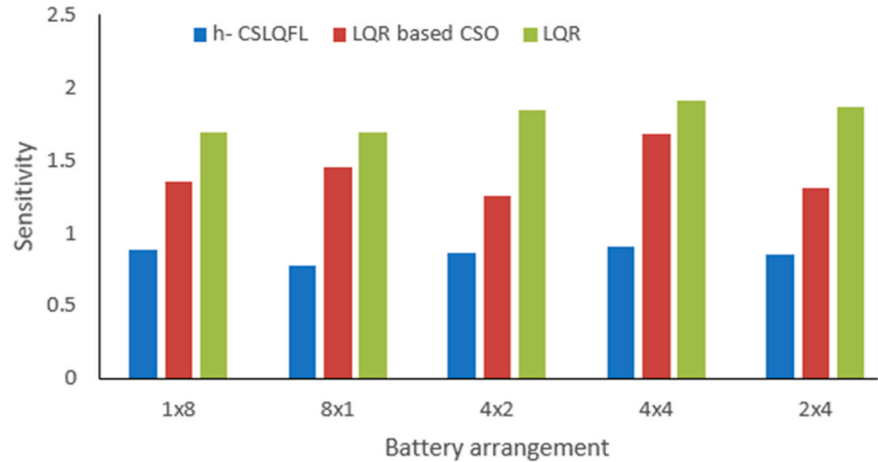


Fig. 10. Comparison of sensitivity for various battery configurations.

The continuous charging and discharging of the battery may arise the need to do sensitivity analysis. The mathematical expression of sensitivity is mentioned in Eq. 49

$$\text{Sensitivity} = \frac{\text{SOC}(t) - \text{SOC}(t-1)}{\text{SOC}(t)} \quad (49)$$

The comparison of sensitivity with various methods for the different arrangement of battery is given in Table 6 and its graphical analysis is shown in Fig. 10.

It is observed that least sensitivity has been found with h-CSLQFL controller in comparison with LQR based CSO and LQR. The modelling of electric vehicles has been designed in an appropriate way, which further measures the performance parameters like battery size, battery cost, risk factor, and sensitivity. The battery size and battery cost for the different battery configuration of the arrangement have been measured which is shown in Table 3 and Fig. 7 & Table 4 and Fig. 8. It is observed that least battery size and battery cost has been attained with hybrid intelligent controller in comparison with LQR based CSO and LQR. Subsequently, risk factor has been analysed with various techniques under different loading conditions which is shown in Table 5 and Fig. 9. Similar dominance of hybrid intelligent controller over techniques have been assessed for the estimation of least risk factor. Further, sensitivity analysis has been done for the different battery arrangement with various techniques which is shown in Table 6 and Fig. 10. It is observed that minimum sensitivity has been attained with hybrid intelligent controller over techniques.

8. Validation of proposed control methodology with experimental set up

The validation of the proposed method with the experimental set up

demonstrated in two ways as, modelling and feasible analysis:

8.1. Modelling and design of experimental set up

The proposed hybrid intelligent controller has also validated on the hardware set up. The hardware or experimental set up consist of solar 2 kW solar PV array which is used charge the lithium-ion batteries of electric vehicle arranged in various configuration. It begins with the transfer of power from solar PV array to the batteries of electric vehicles via DC-DC converter. There is total 8 batteries have been taken and with various possible configuration (1×8, 8×1, 2×4, 4×2, 4×4,) but the configuration of 4×4 has been taken which is giving the best optimal result. The block diagram of the experimental set can be represented as shown in Fig. 11 in which output of 2 KW solar PV array is given to two parallelly connected DC-DC converters (boost converter and buck converter) via switch S₁ and S₂. Further the output of DC-DC converters is connected to lithium-ion batteries. The real time model of experimental set up is shown in Fig. 12. The controlling of boost converter has been assessed with PIC micro-controller circuit while the controlling of buck converter has been done with STEM32F407 micro-controller. Both these micro controllers of hardware set up are interfaced with simulation in which proposed h-CSLQFL controller has been developed. In the simulation model, a multi-objective function in terms of battery cost, battery size, SOC has been developed which is trained by using proposed h-CSLQFL controller. Further, the pulses given for the switching of two DC-DC converter has been assist by using hybrid intelligent h-CSLQFL controller.

The rating of different components used in the hardware set up has been mentioned in Table 7. From the Fig. 12, it can be seen that model for hybrid intelligent h-CSLQFL controller has been designed through Simulation model in desktop system. Further, software programming

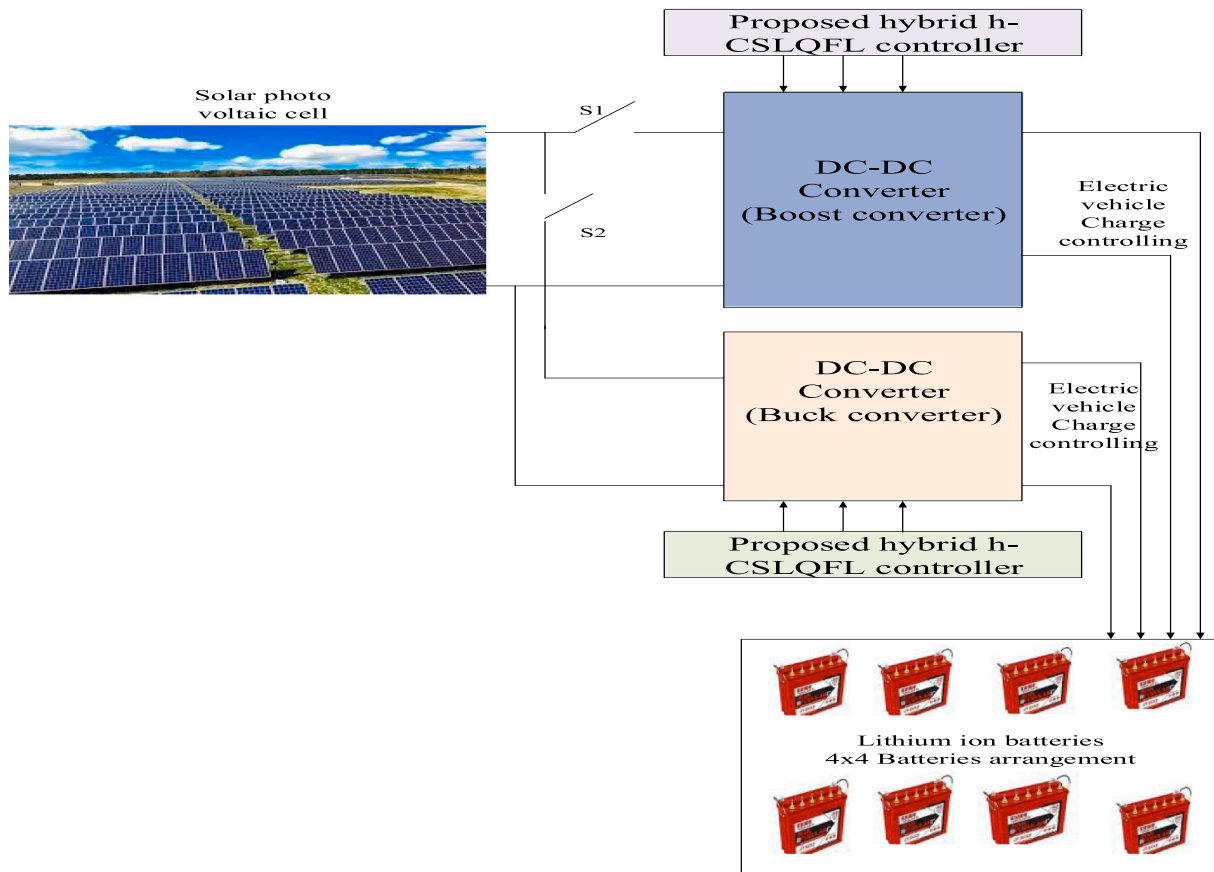


Fig. 11. Structure of the block diagram for the experimental set up.

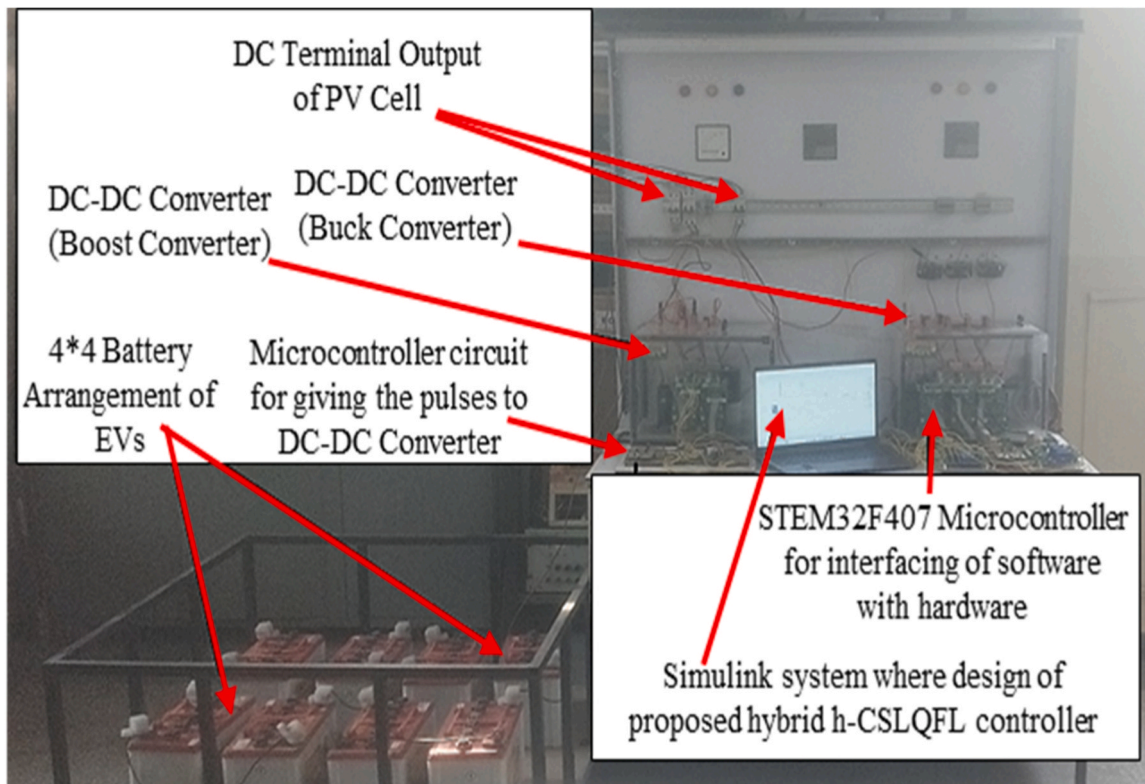


Fig. 12. Experimental set up for electric vehicle charging based on DC-DC converter controlling.

Table 7

Parameters of the experimental set up.

MODULE	Parameter	Value
Solar Module	Plant Power Rating	2 kW
	Power Rating per Panel	350 W
	Voltage Rating	45 V
	Current Rating	9.1
	No. of Panel	6
	Series Combination	6
DC- DC converter	Parallel Combination	1
	Power Rating	3 kW
	Inductance	3 mH
	Capacitor	6μF
	Input Voltage	96 v
	Output Voltage	230 V
	Duty Cycle	0.45
	Operating Frequency	10 kHz
Controller	STM32	STM32F407 Discovery
Battery Bank	No. of Batteries	8
	Per Battery voltage	12 V
	Per Battery current	40Ah
	Connection	Series
	Combined Voltage	96 V

has been interfaced with hardware set up through STM32F407 controller. The hybrid intelligent controller has been used to give the switching pulses to DC-DC converter. The proposed hybrid intelligent h-CSLQFL controller depends on battery size and battery cost which further readjusted with its optimal value for switching of DC-DC converters. As SOC characteristics of battery has charging time (T_{on}) and discharging time (T_{off}), which further give switching pulses(α) to DC-DC converter as shown in Eq.49.

$$\alpha = \frac{T_{on}}{T_{on} + T_{off}} \quad (49)$$

The SOC characteristics of electric vehicle battery is shown in Fig. 13

and its mathematical expression is given in Eq.50.

$$SOC(t) = SOC(t-1) - \frac{it}{Q} \quad (50)$$

8.2. Performance analysis of experimental set up

The experimental set up also measures the parameters like battery cost, battery size, and risk factor. The performance parameters have been compared with experimental set up with proposed method and other methods have been shown in Table8, Table9, and Table10

From the Table8, battery size for different battery arrangement with various methods have been shown. It can be drawn that optimal reduced value of battery size attained with proposed h-CSLQFL controller is very close to hardware set up in comparison to other methods.

From the Table9, electric vehicle battery cost for different battery arrangement with various methods have been shown. It can be drawn that optimal minimum value of battery cost attained with proposed h-CSLQFL controller is very close to hardware set up in comparison to other methods.

From the Table10, risk factor of battery for different battery arrangement with various methods have been shown. It can be drawn that optimal minimum value of risk factor attained with proposed h-CSLQFL controller is very close to hardware set up in comparison to other methods. The significant conclusion extracted from the overall performance analysis is that all the performance parameters attained with experimental set up are very close to proposed hybrid intelligent controller which shows the validation of proposed h-CSLQFL controller.

9. Conclusions

In order to improve the various performance parameters of the battery of Electric vehicle systems (EVs), this paper presents the design and control of the h-CSLQFL controller. The hybrid h-CSLQFL controller has been framed by combining the three techniques: Cuckoo search

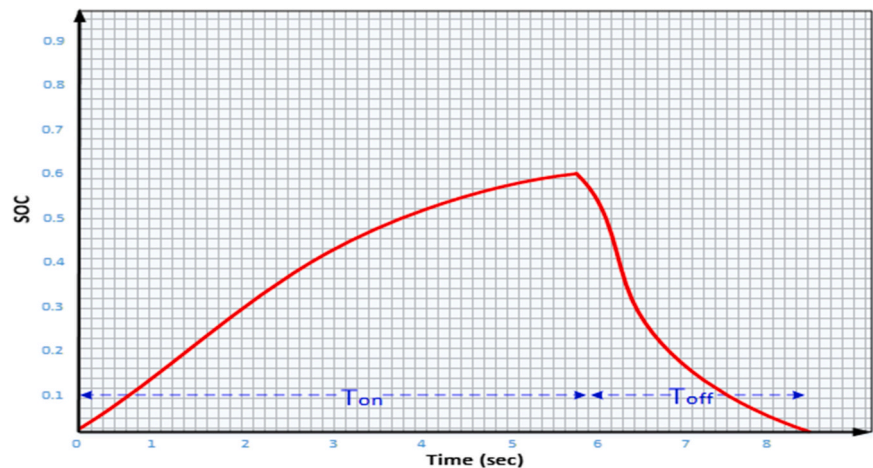


Fig. 13. SOC characteristics of electric vehicle battery with respect to time.

Table 8
Comparison of Battery Size $f(Bs''(t)'',N'')/Kwh$ for different Arrangement of Battery with experimental set up and different control methods.

<i>n</i>	Experimental set up	<i>h- CSLQFL</i>	<i>LQR based CSO</i>	<i>LQR</i>
1×8	39.9	39.8	41	43.7
8×1	39.4	39.2	40.4	43.1
4×2	40.5	40.1	41.3	44
4×4	39.1	38.9	40.1	42.8
2×4	40.1	39.9	41.1	43.8

Table 9
Comparison of Electric Vehicle battery cost ($BP(t)$) in Dollar/ Kwh for different Arrangement of Battery with experimental set up and different control methods.

<i>n</i>	Experimental set up	<i>h- CSLQFL</i>	<i>LQR based CSO</i>	<i>LQR</i>
1×8	1.25	1.22	1.51	1.91
8×1	0.99	0.9	1.12	1.52
4×2	1.05	1.01	1.21	1.61
4×4	0.86	0.84	0.99	1.39
2×4	1.26	1.21	1.35	1.75

Table 10
Comparison of risk factor for different loading conditions with experimental set up and different control methods.

Different loading condition	<i>n</i>	Experimental set up	<i>h- CSLQFL</i>	<i>LQR based CSO</i>	<i>LQR</i>
25 % of full load	1×8	0.79	0.78	0.81	0.85
40 % of full load	8×1	0.69	0.68	0.7	0.72
60 % of full load	4×2	0.68	0.66	0.68	0.71
80 % of full load	4×4	0.35	0.32	0.45	0.51
full load	2×4	0.49	0.48	0.51	0.56

optimization (CSO), linear quadratic regulator (LQR), and fuzzy logic controller (FLC). The major problems with EVs are higher battery cost and large battery size. In addition to this, high risk factor and poor sensitivity are also more the concern factor. A strategy for resolving the various issues has been planned by framing a multi-objective function that is controlled by an h-CSLQFL controller in order to enhancing the various performance parameters. It begins with the development of an EV mathematical model, in which the input parameters are supply voltage and current and the output parameters are speed and torque. After that, a LQR controller has been designed and implemented to examine EV performance parameters. Following that, a closed-loop CSO controller in assist with a LQR controller has been developed. In the end, an LQR-based CSO and LQR controller are used to govern the multi-

objective function. In compared to the LQR based CSO and LQR controller, it has been concluded that the h-CSLQFL controller achieves the least optimal economic cost and minimum size of an EV battery. The least risk factor and improved sensitivity have also attained with hybrid controller which ensures the safety of EV. The proposed h-CSLQFL controller has been verified using a hardware configuration.

Data availability statement

Not applicable.

CRedit authorship contribution statement

ATIF IQBAL: Validation, Supervision, Project administration, Formal analysis. **Talal Alharbi:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis. **Abhinav Saxena:** Writing – original draft, Methodology, Investigation. **Rajat Kumar:** Writing – review & editing, Software, Resources. **Mohammad Amir:** Writing – original draft, Software, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Institutional Review Board Statement

This article does not contain any studies with human participants/ animals performed by any of the authors.

Informed Consent Statement

Not applicable.

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